

AAYUSH PANDA

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TECHNICAL SKILLS

Programming Languages: C, C++, Rust, Python, Java, TypeScript, SQL, Haskell, Scheme, Bash, HTML/CSS, x86 assembly
Machine Learning: Transformers, RNNs, flow-matching, VAEs, PyTorch, Tensorflow, Sklearn, Numpy
Databases: SQL dbs, Firestore, Neo4J, Chroma

EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours Co-op

GPA: 3.90/4.00

Sep. 2024 – Apr. 2028

EXPERIENCE

ML & NLP Engineering Intern

Sep. 2025 – Apr. 2026

Huawei Canada

- Developed algorithms for knowledge graph generation from natural language documents, with named entity linking
- Improved performance of Neo4j knowledge graph updates using Merkle trees. Reduced latency by **95%**
- Developed document content indexing for AI text editor with $F_1 = 0.95$ chunk retrieval in large documents (100+ pages)
- Adapted a hybrid ASR model architecture to work with natural language context to guide transcription, reduced WER by $\sim 5\%$
- Developed agents and encoder-only models for voice/drawn gesture productivity software use (**10 million users**)
- Developed iteratively self-improving coding agents, with outputs being preferred to production code in **90%** of out-of-dist. tasks

Laboratory Assistant

Apr. 2025 – Aug. 2025

University of Waterloo Multi-Sensory Brain and Cognition Lab (MBC)

- Research on how batsmen's eyes track baseballs, measuring visual perception bandwidth, and correlating with batting ability
- Developed hybrid/synthetic data pipelines to train a model to track baseballs in captured video with **97%** success rate
- Developed correspondence algorithm to synchronise data from high speed head mounted camera and eye tracking data

Undergraduate Research Assistant

Apr. 2025 – Aug. 2025

University of Waterloo Data Privacy Lab

- Research on private record linkage protocols using locality sensitive hashing, supervised by Dr. Florian Kerschbaum
- Worked on alternatives to frequency-smoothing as in Wei and Kerschbaum 2023 offering **5 – 10x** speedup on testcases
- Developed novel private set intersection algorithm using k-d cuckoo hashtables, reducing runtime from $O(n^2)$ to $O(n)$

Software Engineering Intern (Contract)

Mar. 2025 – Apr. 2025

Toma (YC W25)

- Developed automated lead qualification service that evaluates dealerships by calling them with a conversational voice AI.
- Improved observability with unified call dashboard with synchronised audio, text transcripts and interactive server logs

PROJECTS

🔗 Concord — Rust, Vulkan, CUDA, HLSL

- Cross-platform runtime in Rust for accelerated computation, with a unified ISA and memory model for CPU/GPU operations.
- Architected bytecode ISA and VM runtime with native cooperative multitasking and FFI
- Developed highly modular compilation toolchain with inline bytecode support to allow for elegant native extensions
- Created CLI interactive debugger with VM state inspection and variable watching

🔗 Semantify — Python, React, FastAPI, Sentence Transformers, UMAP, Ollama

- Built a system that semantically organises files into a directory structure (1000+files/s) using embedding models and custom hierarchical clustering and subtree merging algorithms
- Developed an interactive visualization of document embeddings to allow intuitive semantic exploration
- Implemented a RAG-powered chat interface to intelligently query uploaded documents with source citation

🔗 Phased Array SONAR — C++, Python, Xtensa LX6 microprocessor, AVR RISC processors

- Engineered a sub-degree precision beam-steering phased SONAR array with real-time radar-style display on oscilloscopes.
- Designed an acoustic waveguide to suppress grating lobes, enhancing beam directivity.
- Built a phased array simulator suite to visualize beamforming, steering, and beam focussing behaviors in 2D.

PATENTS

CA 3222437: Device for redirection of optical beams using virtual gratings generated by stationary waves.

CA 3119717: Compliant mechanism for operating flight control surfaces of a remotely piloted aircraft.

AWARDS

University of Waterloo President's Research Award

Jane Street Estimathon @ UWaterloo (2024): First place

Hack the North 2022: Winner (out of 829 participants)

PicoCTF: 2nd place in Canada, 14th (top 0.001%) globally